ANSWERS

Chapter 7

EXERCISE 7.1

(1) (i) 21 m/s (ii) 15 m/s and 27 m/s

(2) (2)(i) 5sec (ii) 128 ft/s (iii)160 ft/s

(3) (i) 1,2 sec (ii) 34 m (iii) $-6\,m/s^2$, $6\,m/s^2$

(4) 75 units (5) $\frac{1}{2}$ kg/m, $\frac{1}{6}$ kg/m

(6) 20π sq.cm/s (7) 2π km/s (8) $\frac{9}{10\pi}$ m/min

(9) (i) $\frac{-8}{3}$ m/s (ii) 26.83 sq.m/sec (10) 70 km/hr.

EXERCISE 7.2

(1) (i) 7 (ii) ∞ (2) $(1,0)$

(3) $(0,3)$ and $(4,-25)$ (4) $(2,-1)$ and $(-2,1)$

(5) (i) $2x+y=2\,;x-2y=1$ (ii) $2x-y=-2\,;x+2y=4$

(iii) $x-y=0\,;x+y=\pi$ (iv) $4x+2y=5\;;2x-4y=-5$

(6) $12x-y=15\,;12x-y=-17$ (7) $x+2y=7\,;x+2y=-1$

(8) $(2\cos t)x+(7\sin t)y=14\,;\ (7\sin t)x-(2\cos t)y=45\sin t\cos t$ (9) $\tan^{-1}(3)$

EXERCISE 7.3

(1) (i) not continuous at $x=0$ (ii) not continuous at $x=\frac{\pi}{2}$ (iii) $f(2)\neq f(7)$

(2) (i) $\frac{1}{2}$ (ii) $-2+2\sqrt{2}$ (iii) $\frac{9}{4}$

(3) (i) not continuous at $x=0$ (ii) not differentiable at $x=\frac{-1}{3}$

(4) (i) $\pm\frac{2}{\sqrt{3}}$ (ii) 7

(6) 320 km (8) No. Since $f'(x)$ cannot be 2.5 at any point in $(0,2)$.

EXERCISE 7.4

(1) (i) $e^x=1+\frac{x}{\lfloor 1}+\frac{x^2}{\lfloor 2}+\cdots$ (ii) $\sin x=x-\frac{x^3}{\lfloor 3}+\frac{x^5}{\lfloor 5}-\frac{x^7}{\lfloor 7}+\cdots$

(iii) $\cos x=1-\frac{x^2}{\lfloor 2}+\frac{x^4}{\lfloor 4}-\frac{x^6}{\lfloor 6}+\cdots$

(iv) $\log(1-x)=-\left(x+\frac{x^2}{2}+\frac{x^3}{3}+\frac{x^4}{4}+\cdots\right)$

(v) $\tan^{-1}(x)=x-\frac{x^3}{3}+\frac{x^5}{5}-\frac{x^7}{7}+\cdots$ (vi) $\cos^2 x=1-\frac{2x^2}{\lfloor 2}+\frac{2^3x^4}{\lfloor 4}-\frac{2^5x^6}{\lfloor 6}+\cdots$

(2) $\log x=(x-1)-\frac{1}{2}(x-1)^2+\frac{1}{3}(x-1)^3-\frac{1}{4}(x-1)^4+\cdots$

(3) $\frac{\sqrt{2}}{2}\left(1+\frac{1}{\lfloor 1}\left(x-\frac{\pi}{4}\right)-\frac{1}{\lfloor 2}\left(x-\frac{\pi}{4}\right)^2-\frac{1}{\lfloor 3}\left(x-\frac{\pi}{4}\right)^3+\cdots\right)$

(4) $f(x)=-(x-1)+(x-1)^2$

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